

Geneva, April 7, 2026

Dear Editorial Board Members,

Please find enclosed a manuscript entitled “**Mindfulness training impacts brain network dynamics linked to stress response in young adolescents.**” We are submitting it for exclusive consideration of publication as a research article in the Human Brain Mapping journal.

This paper uses recent advances in network neuroscience and neuroimaging to study how brain activity changes during rest after acute psychosocial stress. Specifically, it examines whether mindfulness trait is linked to patterns of dynamic brain connectivity and how mindfulness-based interventions affect brain responses to neutral, negative, and positive stress conditions. This work moves beyond the traditional focus on static brain responses to stress by examining how intrinsic brain networks change over time.

Using a novel approach, we characterized the spatial and temporal patterns of brain-wide activity and connectivity with functional co-activation patterns (CAPs) in resting-state fMRI following acute psychosocial stress of varying affective valence (neutral, negative, and positive). This approach showed that mindfulness-based interventions (MBI) reduced transitions into a specific brain state involving connectivity between the posterior cingulate cortex (PCC) and regions typically associated with the dorsal medial default network (CAP_{DN-DM}). Variability in this CAP_{DN-DM} was strongly linked to clinical symptoms, with mindfulness trait serving as a causal mediator. These effects were robust to unmeasured confounding and were not explained by pubertal maturation.

By linking CAP dynamics to mindfulness trait and stress-related symptoms, these findings suggest that CAPs may serve as promising neural markers to guide intervention strategies such as MBI. They also add to growing evidence that mindfulness may shape brain function in ways relevant to stress resilience and vulnerability, beyond traditional rs-fMRI studies focused on global abnormalities associated with disease.

We believe these results will be of high interest to researchers and clinicians in neuroscience and psychiatry, as well as psychologists, neuroimaging scientists, and signal processing engineers interested in brain dynamics, network neuroscience, emotion regulation, and fMRI methodology.

Yours sincerely,



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